

Project A4: ML+SDCC 2

- Decentralized federated learning system
 - Team: 2 students per team, max 4 teams
 - Goal: fully decentralized federated learning system for neural networks
 - K clients, each with local private data
 - Any client can initiate training and select aggregation strategy (e.g., FedAvg)
 - Evaluation on LEAF datasets with multi-user data per client <https://leaf.cmu.edu/>
 - Analysis of scalability: accuracy and training time vs. number of clients
 - Be careful to communication overhead
 - Limited centralization: allowed only for coordination tasks (e.g., service registry)

Projects B: summary

- B1: Distributed service registry
- B2: Carbon and/or energy aware load balancer
- B3: Gossip-based distributed data aggregation
- B4: Microservice application of your choice
- For all B projects:
 - Go as programming language
 - 1 student per team, max 4 students per project